Guide to the Algorithmic Redistricting Portfolio

An 11-Paper Research Program

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Overview

Congressional redistricting in the United States suffers a legitimacy crisis: partisan mapmakers manipulate district boundaries to predetermine election outcomes, eroding public confidence in representative democracy. This portfolio demonstrates that **recursive graph bisection**—a computational method used for supercomputer workload distribution—provides an algorithmic alternative that *cannot gerrymander because it cannot access partisan data*.

Applied to all 50 states across three census decades (2000/2010/2020), this method produces 435 congressional districts with near-perfect population equality, geographic contiguity, and geometric compactness. Algorithmic plans create **69 more majority-minority districts** than enacted maps, exceeding Voting Rights Act requirements without explicit racial targeting. We identify a **42% demographic threshold** determining where proportional minority representation becomes geographically feasible.

The Big Picture: Four Contributions

- **Philosophical**: *Impossibility defense*—algorithm cannot see partisan data, therefore cannot gerrymander
- **Technical**: Edge-weighted graph partitioning achieves +56% compactness improvement while enabling VRA compliance
- **Empirical**: National-scale validation across 50 states × 3 census decades (150 redistricting scenarios)
- **Legal**: 42% state minority threshold determines VRA feasibility; algorithmic surplus of +69 majority-minority districts

Portfolio Architecture

The 11-paper portfolio follows a hierarchical structure:

- **Foundation** (Paper 01): Recursive bisection method and impossibility defense
- **Primary Extensions** (Papers 02-03): Compactness optimization and demographic awareness
- **Systematic Investigations** (Papers 04-10): Seven papers exploring properties, limits, and alternatives

Paper Summaries

All 11 papers are complete and ready for submission.

#	Paper	Key Finding
01	Recursive Bisection	Impossibility defense: algorithm can't see partisan data
02	Edge-Weighted Bisection	+56% compactness improvement with edge-weighting
03	VRA Compliance	+69 MM district surplus over enacted plans
04	Threshold Analysis	42% state minority threshold for VRA feasibility
05	Cross-Census Validation	3.2× variance geography > time (algorithm stable)
06	Compactness Tradeoff	Non-MM districts gain +7.5% compactness (not sacrifice)
07	Temporal Stability	+14pt tract retention: recursive > n-way stability
08	Multi vs Edge	Edge-weighting beats multi-constraint optimization
09	Adaptive Bisection	Tree structure irrelevant with $\alpha \geq 5$ edge-weighting
10	N-Way vs Recursive	Statistical equivalence: both methods viable
11	Efficiency Gap Analysis	62% partisan bias reduction compared to enacted plans

Recommended Reading Order

For Political Scientists

1. **Paper 01** (Recursive Bisection): Impossibility defense, philosophical foundation, Huntington-Hill precedent
2. **Paper 03** (VRA Compliance): +69 MM district surplus, demographic representation
3. **Paper 04** (Threshold Analysis): 42% finding, legal implications, Gingles test

For Computer Scientists

1. **Paper 02** (Edge-Weighted Bisection): Algorithm innovation, compactness optimization
2. **Paper 08** (Multi vs Edge): Constraint conflict theory, why single-objective wins
3. **Paper 09** (Adaptive Bisection): Parameter sensitivity analysis

For Legal Scholars

1. **Paper 01** (Recursive Bisection): *Rucho* implications, mathematical governance
2. **Paper 03** (VRA Compliance): Section 2 feasibility without racial targeting
3. **Paper 04** (Threshold Analysis): 42% threshold, Gingles preconditions
4. **Paper 06** (Compactness Tradeoff): VRA-compactness myths debunked

For Practitioners (Redistricting Commissions)

1. **Paper 01** (Recursive Bisection): Core method overview
2. **Paper 02** (Edge-Weighted Bisection): Implementation details for compact districts
3. **Paper 03** (VRA Compliance): Approach for achieving VRA objectives
4. Skim others as needed for specific questions

Glossary of Key Terms

Census tracts
Geographic units used as atomic partitioning units (∼4,000 residents)

METIS Multilevel graph partitioning library by Karypis & Kumar

Edge-weighted partitioning
: Assigning higher costs to edges between similar nodes (making them expensive to cut), encouraging spatially compact districts

Majority-minority districts
: Districts where minority voters $\geq$50% of voting-age population

Polsby-Popper
: Compactness measure: $4\pi \times$ Area/Perimeter^2, range [0,1]

Recursive bisection
: Hierarchical method splitting graph into two parts repeatedly

Impossibility defense
: Algorithm cannot gerrymander because it cannot access partisan data

VRA
: Voting Rights Act (1965)—federal law prohibiting minority vote dilution

Code and Data

- **Replication repository**: [Available upon request or publication]
- **Census data sources**: U.S. Census Bureau redistricting files (2000/2010/2020)
- **METIS library**: `http://glaros.dtc.umn.edu/gkhome/metis/metis/overview`
- **Contact**: giovanni.deluca@example.com

Citation Guide

Publication status: All 10 papers are complete and ready for submission (as of February 2026). Preprints will be available on arXiv upon initial submission.

For the entire research program:

> Della-Libera, G. (2026). Recursive bisection for congressional redistricting: A 10-paper research program. Preprint series.

For specific papers (example):

> Della-Libera, G. (2026). Recursive bisection for congressional redistricting. Preprint, submitted to *American Political Science Review*.

About this guide: This 2-page overview introduces readers to the 10-paper algorithmic redistricting portfolio. For detailed findings, see individual papers. For synthesis across papers, see Paper 00 (Synthesis Metapaper for *Science*).